Relations

Chapter 9

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Chapter Summary

9.1 Relations and Their Properties

9.2 *n*-ary Relations and Their Applications

9.3 Representing Relations

9.4 Closures of Relations

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Relations and Their Properties

Section 9.1

9.1.1 Introduction

To express a relationship between elements of two sets is to use ordered pairs made up of two related elements.For this, sets of ordered pairs are called binary relations.

DEFINITION 1: Let A and B be sets. A binary relation (R) from A to B is a subset of A X B.

$$R \subseteq A \times B.$$

In other words, a binary relation from A to B is a set R of ordered pairs, where the first element of each ordered pair comes from A and the second element comes from B.

- aRb to denote that (a, b) ∈ R
- a$\not$R b to denote that (a, b) ∉ R.

Binary relations

Binary relations represent relationships between the elements of two sets.

We will introduce n-ary relations, which express relationships among elements of more than two sets, later in this chapter.

We will omit the word binary when there is no danger of confusion.

Binary Relations

Let A be the set of students in school, and let B be the set of courses. Let R be the relation that consists of those pairs (a, b), where a is a student enrolled in course b.

For instance, if Jason and Deborah are enrolled in CS518,

the pairs (Jason, CS518) and (Deborah, CS518) belong to R.

If Jason is also enrolled in CS510,

then the pair (Jason, CS510) is also in R.

However, if Deborah Sherman is not enrolled in CS510,

then the pair (Deborah, CS510) is not in R.

Note that if a student is not currently enrolled in any courses there will be no pairs in R that have this student as the first element.

Similarly, if a course is not currently being offered there will be no pairs in R that have this course as their second element.

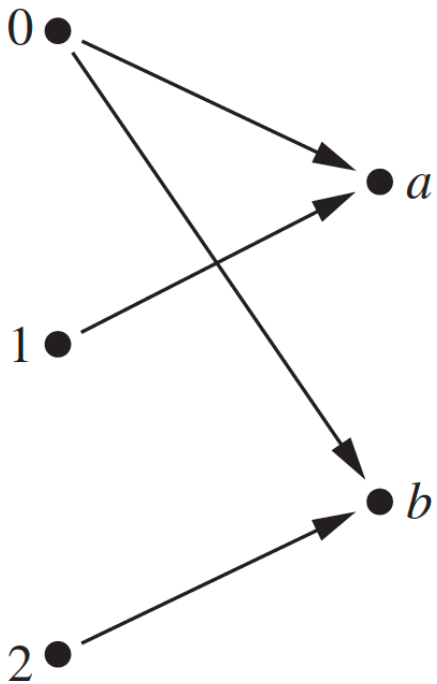
Binary Relations

Let A = {0, 1, 2} and B = {a, b}.

{(0, a), (0, b), (1, a), (2, b)} is a relation from A to B.

This means, for instance, that 0Ra, but that $1 \not R\ b$.

Relations can be represented graphically or using a table.



R	a	b
0	×	×
1	×	
2		×

using arrows to represent ordered pairs.

9.1.2 Functions as Relations

Recall that a function f from a set A to a set B (as defined in Section 2.3) assigns exactly one element of B to each element of A.

The graph of f is the set of ordered pairs (a, b) such that $b = f(a)$.

Because the graph of f is a subset of A X B, it is a relation from A to B.

$$(a, f(a)) \text{ for } a \in A$$

Moreover, the graph of a function has the property that every element of A is the first element of exactly one ordered pair of the graph.

Relations are a generalization of graphs of functions; they can be used to express a much wider class of relationships between sets.

9.1.3 Relations on a Set

Relations from a set A to itself are of special interest.

DEFINITION 2: A relation on a set A is a relation from A to A.

In other words, a relation on a set A is a subset of A X A.

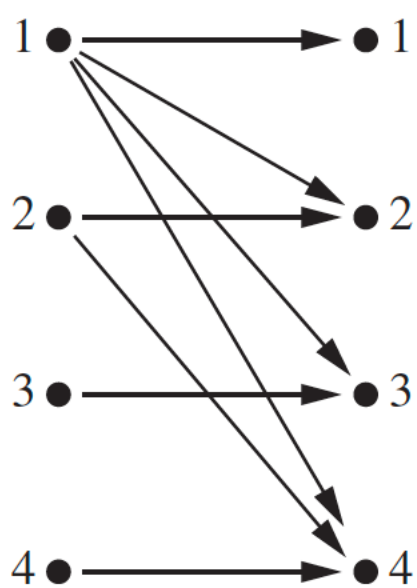
EXAMPLE: $A = \{a,b,c\}$.

Then $R = \{(a,a),(a,b), (a,c)\}$ is a relation on A.

Binary Relations on a Set[1]

Let A be the set {1, 2, 3, 4}. Which ordered pairs are in the relation R = {(a, b) | a divides b}?

R = {(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)}.



R	1	2	3	4
1	×	×	×	×
2		×		×
3			×	
4				×

Binary Relations on a Set$_2$

How many relations are there on a set *A with n elements*?

Solution:

A relation on a set A is a subset of A X A.

Because A X A has n^2 elements when A has n elements, and a set with m elements has 2^m subsets (the cardinality of the power set), there are 2^{n^2} subsets of A X A.

Thus, there are 2^{n^2} relations on a set with n elements.

For EXAMPLE, there are $2^{3^2} = 2^9 = 512$ relations on the set {a, b, c}.

Binary Relations on a Set 3

EXAMPLE: Consider these relations on the set of integers:

$R_1 = \{(a, b) | a \leq b\}$,
$R_2 = \{(a, b) | a > b\}$,
$R_3 = \{(a, b) | a = b \text{ or } a = -b\}$,
$R_4 = \{(a, b) | a = b\}$,
$R_5 = \{(a, b) | a = b + 1\}$,
$R_6 = \{(a, b) | a + b \leq 3\}$.

Which of these relations contain each of the pairs (1,1), (1, 2), (2, 1), (1, −1), and (2, 2)?

(1,1) is in R_1, R_3, R_4 , and R_6

(1,2) is in R_1 and R_6

(2,1) is in R_2, R_5, and R_6

(1, −1) is in R_2, R_3, and R_6

(2,2) is in R_1, R_3, and R_4.

Note that these relations are on an infinite set and each of these relations is an infinite set.

9.1.4 Properties of Relations

Reflexive

Symmetric, Antisymmetric

Transitive

Reflexive Relations

DEFINITION 3: A relation R on a set A is called reflexive if (a, a) ∈ R for every element a ∈ A.

R is *reflexive* iff (*a*,*a*) ∈ *R* for every element *a* ∈ A.

Written symbolically, R is reflexive if and only if

$$\forall x(\mathrm{x} \in U \rightarrow (x, x) \in R)$$

$$\forall \mathrm{a}((\mathrm{a}, \mathrm{a}) \in \mathrm{R})$$

If *A* = ∅ then the empty relation is reflexive vacuously.
That is the empty relation on an empty set is reflexive!

Reflexive Relations

Consider the following relations on {1, 2, 3, 4}:

R1 = {(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)},

R2 = {(1, 1), (1, 2), (2, 1)},

R3 = {(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)},

R4 = {(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)},

R5 = {(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)},

R6 = {(3, 4)}.

Which of these relations are reflexive?

$$\forall x(\mathrm{x} \in U \rightarrow (x, x) \in R)$$

Reflexive Relations

Consider the following relations on {1, 2, 3, 4}:

R1 = {(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)}, ❌

R2 = {(1, 1), (1, 2), (2, 1)}, ❌

R3 = {(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)}, ✔

R4 = {(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)}, ❌

R5 = {(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)}, ✔

R6 = {(3, 4)}. ❌

Which of these relations are reflexive?

$$\forall x(\mathrm{x} \in U \rightarrow (x, x) \in R)$$

The relations R3 and R5 are reflexive because they both contain all pairs of the form (a, a), namely, (1, 1), (2, 2), (3, 3), and (4, 4).
In particular, R1, R2, R4, and R6 are not reflexive because (3, 3) is not in any of these relations.

Reflexive Relations

Which of these relations are reflexive?

✓ $R_1 = \{(a,b) | a \leq b\}$,

❌ $R_2 = \{(a,b) | a > b\}$, $3 \not> 3$

✓ $R_3 = \{(a,b) | a = b \text{ or } a = -b\}$,

✓ $R_4 = \{(a,b) | a = b\}$,

❌ $R_5 = \{(a,b) | a = b + 1\}$, $3 \neq 3+1$

❌ $R_6 = \{(a,b) | a + b \leq 3\}$. $3+3 \not\leq 3$

$\forall x (\mathrm{x} \in U \rightarrow (x,x) \in R)$

Reflexive Relations

Is the "divides" relation on the set of positive integers reflexive?

✔ Because a | a whenever a is a positive integer, the "divides" relation is reflexive.

Is the "divides" relation on the set of all integers reflexive?

❎ The relation is not reflexive because by DEFINITION 0 does not divide 0.)

$$\forall x(\mathrm{x} \in U \rightarrow (x, x) \in R)$$